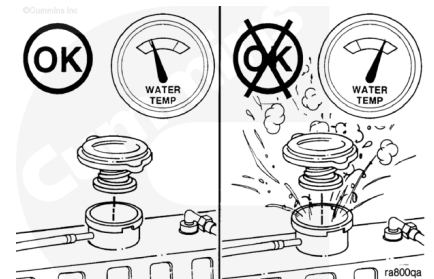


Trainings ▾

Test

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the temperature is below 50°C [120°F] before removing the coolant system pressure cap. Heated coolant spray or steam can cause personal injury.

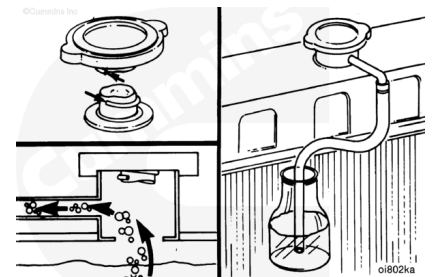


Allow the engine to cool. Remove the radiator cap.

Note : The pressure cap must make a tight seal.

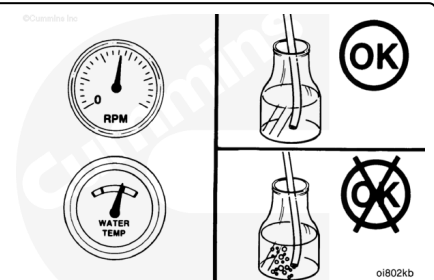
Install a radiator pressure cap that has had the spring and pressure relief valve removed.

Attach a rubber hose to the radiator overflow connection. Put the free end of the hose in a container of water.



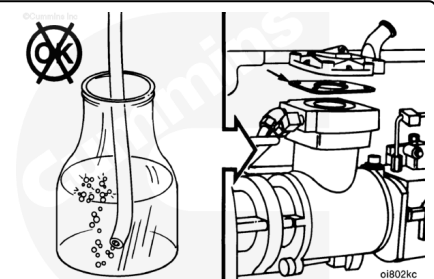
Operate the engine at rated rpm until it reaches a temperature of 80°C [180°F] with the thermostat open.

Check for a continuous flow of air bubbles from the hose in the water container.

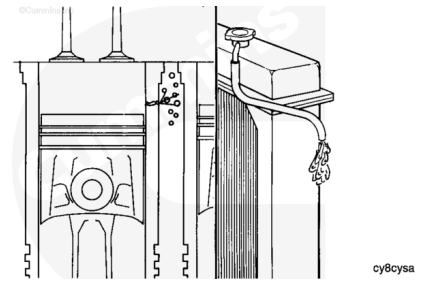


A continuous flow of air bubbles indicates:

- An air compressor cylinder head leak. Inspect the air compressor head and gasket. Refer to Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html>).



- A cylinder head combustion gas leak. Inspect the cylinder head and gasket for damage. Refer to Procedure 002-004 (</qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html>).
- Incorrect cylinder liner protrusion. Refer to Procedure 001-028 (</qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html>).
- Cracked or broken cylinder liners. Refer to Procedure 001-028 (</qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html>).



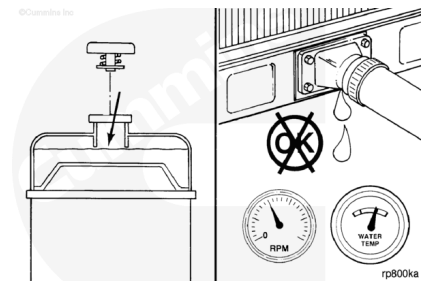
If no air is in the system, do the following.

Remove the test equipment.

Check the coolant level and fill if necessary.

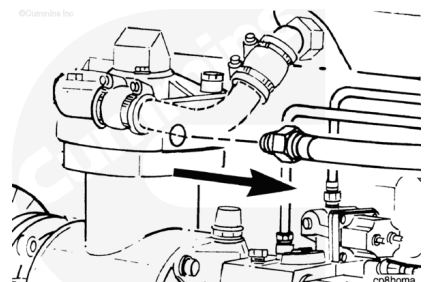
Install the correct radiator pressure cap.

Operate the engine to a temperature of 80°C [180°F], and check for coolant leaks.



⚠ CAUTION ⚠

The air compressor discharge must be disconnected at the compressor. This will allow the compressor to discharge air to the atmosphere to prevent the compressor from overheating during this test.

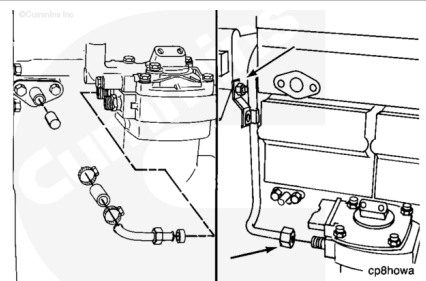


⚠ CAUTION ⚠

Do not operate the engine over 5 minutes with the components isolated from the cooling system. Component damage can result.

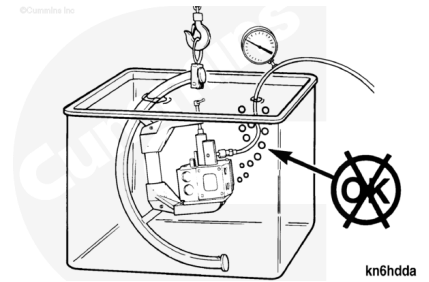
Disconnect and plug the coolant supply and coolant drain lines.

Repeat the test. If no air is found in the cooling system with the air compressor isolated, replace the air compressor head gasket. Refer to Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html>).



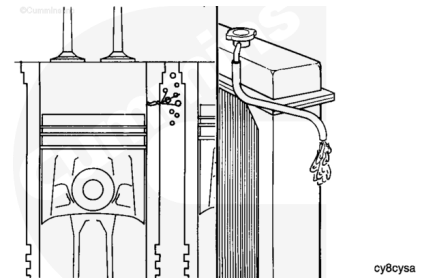
A cracked or defective cylinder head will pressurize the cooling system and force the coolant out the overflow tube.

Check the cylinder head gasket. The part **must** be replaced if defective. Refer to Procedure 002-004 (</qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html>) or 002-021 (</qs3/pubsys2/xml/en/procedures/20/20-002-021.html>).



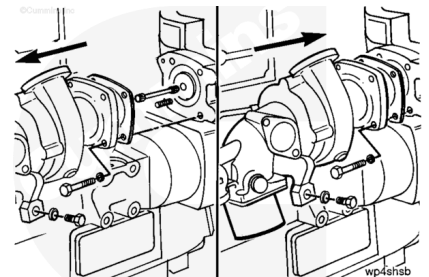
A cracked or porous cylinder liner will pressurize the cooling system and force coolant out the overflow tube.

Check the cylinder liner. The liner **must** be replaced if defective. Refer to Procedure 001-028 (</qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html>).



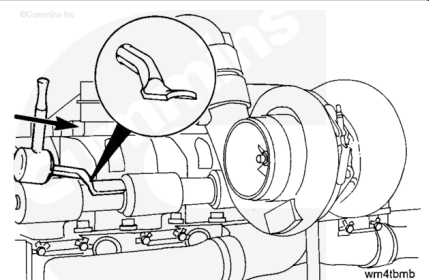
To determine the cylinder liner that is leaking, perform the following:

- Remove the water pump. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>).
- Remove the drive shaft.
- Install the water pump without the drive shaft.



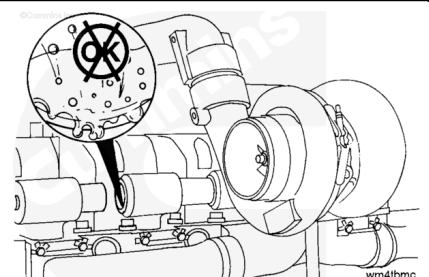
Note : Drain part of the coolant so the coolant level will be level with the water manifold tubes.

Slide the water manifold tubes into the rocker lever housing.

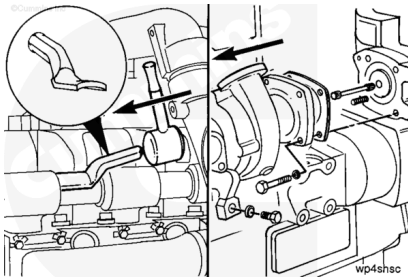


Operate the engine at low idle for a maximum of 4 minutes. Check inside the water passage of each rocker lever housing for air bubbles. Air bubbles in the coolant indicate a porous or cracked cylinder liner.

A defective cylinder liner **must** be replaced. Refer to Procedure 001-028 (</qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html>).

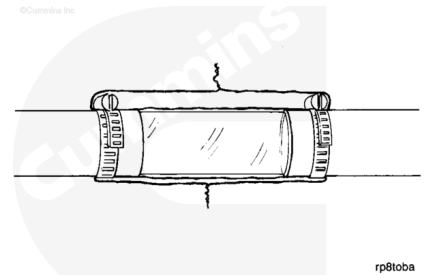


Install the water tubes and water pump drive shaft.



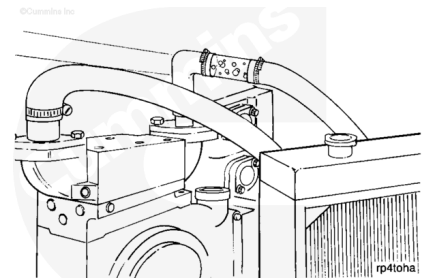
Put a clear Pyrex® tube in the water connection between the engine radiator and the thermostat housing.

Note : If desired, put the Pyrex® tube in the vent line between the thermostat housing and the radiator.



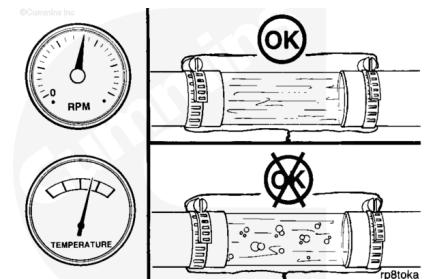
⚠ CAUTION ⚠

Use wire lace to cover the tube and hose. Personal injury can result if the Pyrex® tube separates from the hoses.



Operate the engine at 75 percent of rated rpm until it reaches an operating temperature of 80°C [180°F].

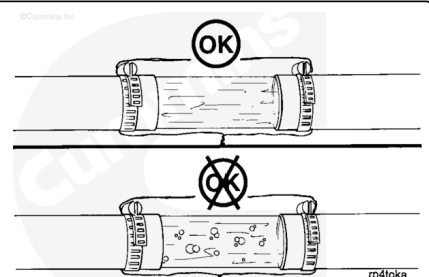
Check for a continuous flow of air bubbles in the hose.



A continuous flow of air bubbles indicates excessive air in the system. Continue to operate the engine for 25 minutes.

If the flow of bubbles does **not** stop, check for the following:

- an air compressor cylinder head leak. Inspect the air compressor head and gasket. Refer to Procedure 012-014 (/qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html).
- a cylinder head combustion gas leak. Inspect the cylinder head and gasket for damage. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html).



- incorrect cylinder liner protrusion. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html).
- cracked or broken cylinder liners. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html).

If no air is found in the system, or if the flow of bubbles stops within 25 minutes, do the following:

Remove the test equipment.

Check the coolant level and fill if necessary.

Install the correct radiator pressure cap.

Operate the engine until it reaches a temperature of 70°C [160°F], and check for coolant leaks.

